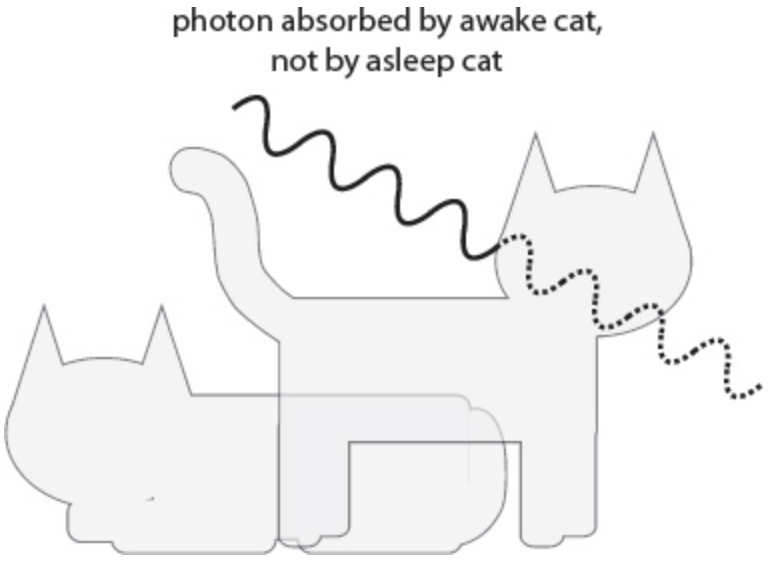


What's special about the "Awake"/"Asleep" basis, in other words, is that the individual states describe well-defined configurations in space. And space is the quantity with respect to which physical interactions are local. A particle can bump into a cat if the particle and the cat come into physical contact. The two parts of the cat wave function, "Awake" and "Asleep," come into contact with different particles in the environment, and therefore branch into different worlds.

This is the basic answer to the question of why we see the particular worlds that we do: the preferred-basis states are those that describe coherent objects in space, because such objects interact consistently with their environments. These are often called *pointer states*, as they are the states in which the pointer of a macroscopic measuring device will indicate a definite value, rather than being in a superposition. The pointer basis is where a well-behaved classical approximation makes sense, and therefore it's that kind of basis that defines emergent worlds. Decoherence is the phenomenon that ultimately links the austere simplicity of Everettian quantum mechanics to the messy particularity of the world we see.

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* Sadly there are competing definitions of the word "emergence," some of which mean almost the opposite of the sense used here. Our definition is sometimes called "weak emergence" in the literature, as opposed to "strong emergence," in which the whole is irreducible to the sum of its parts.